

OIU

Faculty of Pharmacy

Phase 1 / semester 2

Lecture 2

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Main points of the last lecture

In the last lecture: we discuss:

- ❑ Objectives of the course,
- ❑ Definition of pharmacognosy,
- ❑ Present State and scope of Pharmacognosy.

Introduction to pharmacognosy

Outlines

Background

Types and Origin of vegetable drugs

Classification of vegetable drugs

**Binomial system (naming of
crude drugs)**

Objectives of the lecture

By the end of this lecture, the student should be able to:

1. **R**ecognize the history of Pharmacognosy,
2. **E**xplain importance and uses of plants in medicine & pharmacy,
3. **K**now types of drugs derived from plants,
4. **C**lassify vegetable drugs,

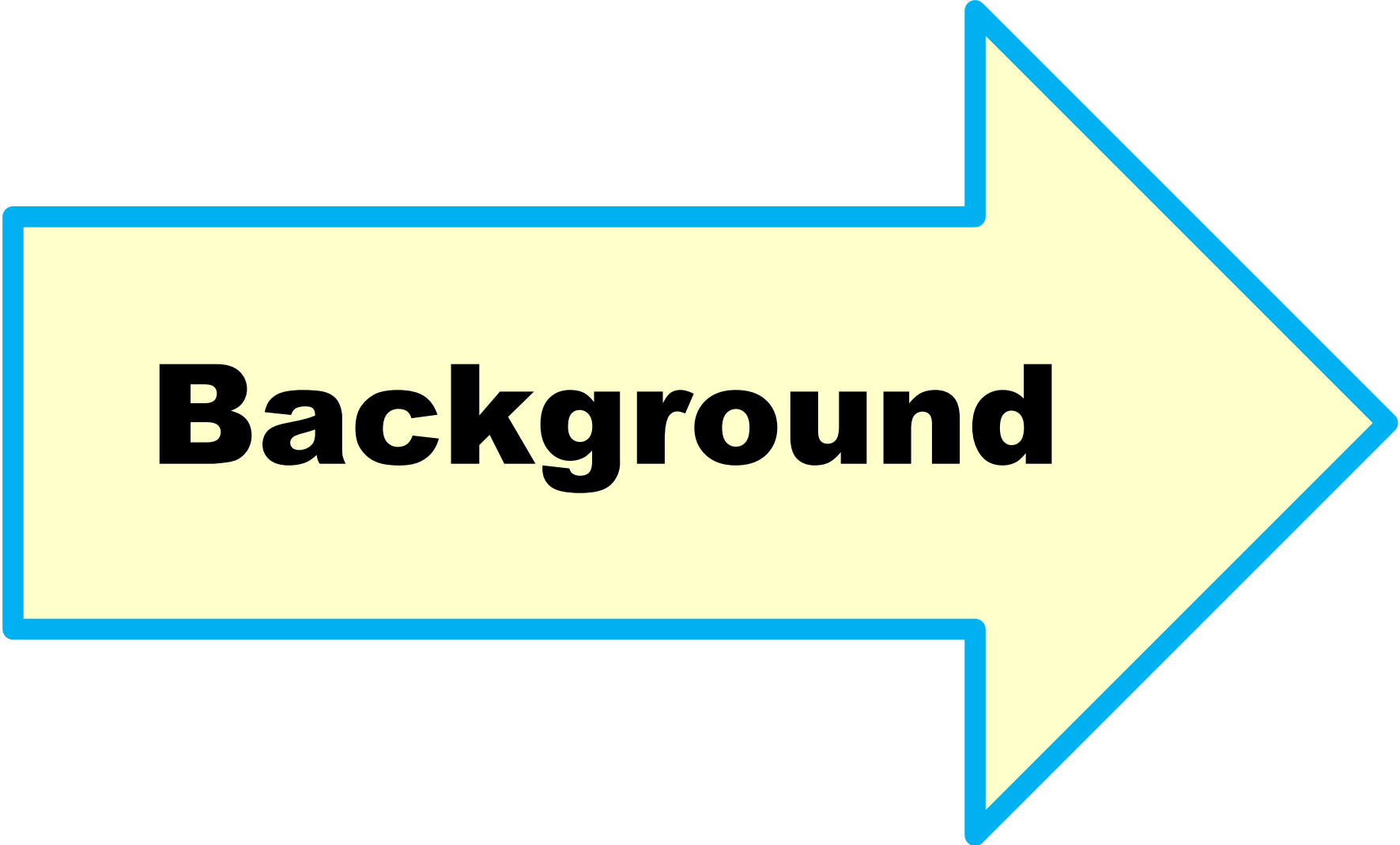
5. **M**ention origins of herbal drugs,
6. **N**ame plants scientifically according to the binomial system.

Background

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Background

Pharmacognosy

The word Pharmacognosy is derived from the Greek :

Pharmakon = a drug or poison &

Gignosis = acquire a knowledge

"the entire knowledge of drug"

History of pharmacognosy

1. Ancient Egyptian period:

- ❑ The ancient Egyptians 3000 B.C. were experts in using drugs for diseases.
- ❑ Crude drugs used included Aloes, Gum, Myrrh, Onion, Anise, Castor, etc...



The first recorded prescriptions were found in Egyptian tombs. E.g Ebers papyrus

2. The Babylonians:

- ❑ The Babylonian medicine was known (Laws of Hamorabi).
- ❑ Mainly of vegetable origin but some are of animal source. Many were known to the ancient Egyptians.

3. Old Indian medicine:

- ❑ Ayurveda contained sacred medicinal plants, e.g Sandal wood, Clove, Caraway, Ginger, Cannabis, Castor oil, Sesame oil, Aloe, etc...
- ❑ The collection was done by religious person
- ❑ The fresh plants were considered to be the most effective.

4. Traditional Chinese Medicine=TCM:

- ❑ Include acupuncture and herbal medicine.
- ❑ Their book includes many recipes for every disease e.g Ginseng, Rhubarb, Opium(very old Chinese drug for diarrhea and dysentery).

5.The Greek and Romans:

- ❑ Used drugs as Mustard and Squill, etc....
- ❑ The Greek Empire was followed by that of the Romans (plants in addition to animal and mineral drugs).
- ❑ Galen was a physician and described "Galenical preparations".

6. Islamic Contribution

- ❑ In Islamic writings were the first beginnings of chemistry (the name derived from Kemia).
- ❑ The Arabs added many new plants to those already known to Greeks & Romans.
- ❑ The first dispensary was in Baghdad, the center of trade in those days,

This dispensary was made of Sandal wood and named "Sandalia".

❑ Rhazey (born in Persia, Alhawi Kabeer) was the director of Baghdad hospital (El-Mansour).

❑ Ibn Al-Baitar (pharmacognosist and botanist) has a book "Jame-ul-Muffradat" contains description of 2000 drugs.

Pharmacopoeia

Pharmacopeia is a book containing direction for: identification, preparation, published by the authority of a government or medical or pharmaceutical society e.g :

- ❑ AHP: American Herbal Pharmacopoeia
- ❑ WHO: World health organization
- ❑ USP, **BP**

Official & Unofficial drugs

- ❑ Official drug: it is one which is listed & described in a book recognized legally as standards (**Pharmacopoeia**).
- ❑ Unofficial drug: those not listed in Pharmacopoeia).

For description of each drug should consider:

1. Origin: biological and geographical.
2. Cultivation and preparation: details of cultivation, collection, drying, packing and others

3. Characters: physical e.g dimensions, colour, odour and taste and histological characters (for powdered form).
4. Constituents and tests: active constituents and also other constituents and chemical identity tests.

5. Adulterants: including materials added and matter which associated with the drug during collection, preparation, packing and transport.

6. Evaluation: of the physical & chemical characters.

7. Uses

Role and Importance of plants in medicine and pharmacy

- ❑ Many drugs were plant based
- ❑ Many drugs were discovered : Morphine &
Quinine

Uses of plants in medicine

- ❑ Source of drug molecules
- ❑ Source of toxic molecules
- ❑ Source of plant drugs
- ❑ Source of lead compounds

Background ✓

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Types and Origin of vegetable drugs

Types of vegetable drugs

1. Herbal drugs derived from specific part of the plant (Organized drugs):

they called:

Herbal medicinal products or

Herbal remedies or

Phytomedicines

A. Entire plants : Mentha,



B. Specific part of plants:

Senna, Clove, Fennel, Linseed, Cinchona,
Liquorice.



2. Natural products or compounds isolated from nature (unorganized):

Opium, Aloes, Tragacanth, Beeswax, Gelatin.



3. Nutraceuticals (functional foods):

Garlic, Spices, Anthocyanin ,Flavonoid containing foods and Carotenoid containing plants



Origin of drugs (or source)

- 1. Natural or biological source:** from the plant (botanical source) or animal (zoological origin).
- 2. Geographical source(Habitat):** the region in which the plant or animal grows. Plants are:
 - *Indigenous: grow in their native countries,
 - *Naturalized: grow in other than native home.

3. The commercial origin:

The commercial origin of a drug refers to its production and its channels of trade.

4. Chemical: (the most important constituents)
e.g. drugs containing volatile oils, glycosides,
alkaloids, saponins, tannins...

5. Pharmacological and therapeutic effects:
(according to the pharmacological action of
most important constituents or their use):
astringent, irritant, drugs affecting GIT,, drugs
used for infectious disease, etc....

Background ✓

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Classification of vegetable drugs

Classification of vegetable drugs

- 1. Alphabetical:** using Latin or English names
- 2. Taxonomic:** botanic-al classification (by plant source): orders, families, genus & species.

3. Morphological :

A) Organized drugs: leaves, flowers, fruits, seeds.

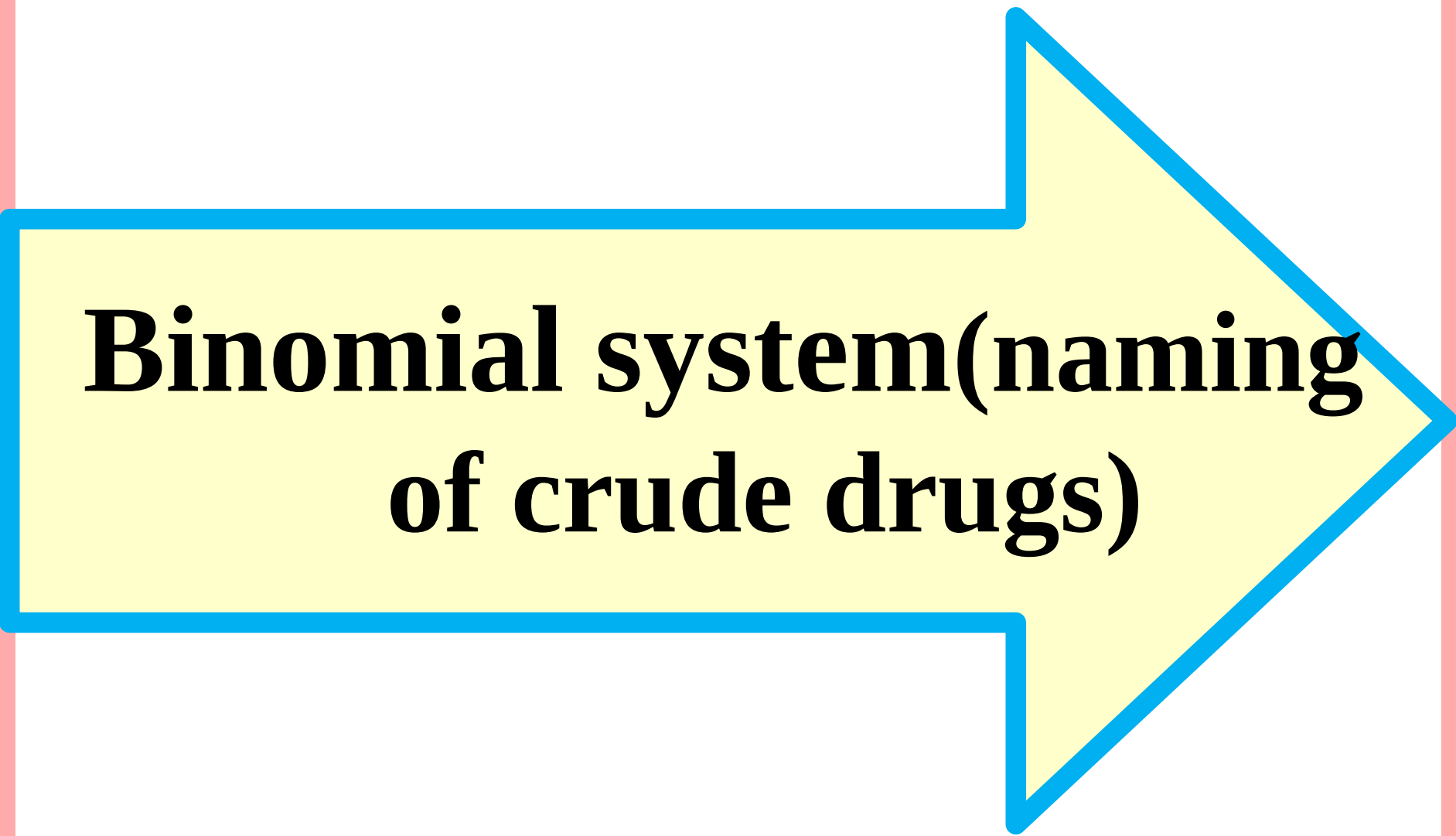
B) Un-organized drugs : dried latex, extracts, gums, resins, oils, fats and waxes.

Background ✓

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**Binomial system (Naming of
crude drugs)**



**Binomial system(naming
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The binomial system

- ❑ Is due to Swedish biologist Linnaeus.
- ❑ The first name: always start w capital letter, denotes the **genus**. The second name start w small letter denoting **species** (can use capital where the species is named by a person name, e.g *Cinchona Ledgeriana*, Charles Ledger brought its seed from Brazil).

The specific name is may chosen to indicate:

1. Some striking characteristics of the plant:

* *Conium maculatum* : maculate = spotted, stem with reddish, spotted patches.

* *Glycyrrhiza glabra* (ous = smooth), fruit is smooth.

* *Hyoscyamus muticus* (short), the plant being short.

* *Atropa belladonna* (beautiful, donna = lady), juice of the berry causes dilatation of the pupils giving a striking appearance).

2. A characteristic colour:

* *Piper nigrum* (black),

* *Digitalis purpurea* (purple).

3. A certain aroma (aromatic plants) :

* *Myristica fragrans* (a fragrant=nice aroma),

* *Caryophyllus aromaticus* (has aroma).

4. Geographical source or history of a drug:

* *Cannabis indica* (growing in India).

5. Pharmaceutical activity or active constituents :

* *Papaver somniferum* (cause somnolence).

* *Strychnos nuxvomica* (cause vomiting).

6. The Generic name also may chosen to indicate certain characters of the plant: :

** Glycyrrhiza*: from glucose(sweet),
riza (root).

Background ✓

**Types and Origin of vegetable
drugs ✓**

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References

❑ Trease and Evans Pharmacognosy. By W.C. Evans. Published by W.B. Saunders 16th, edition Edinburgh London 2009.

❑ Textbook Of Pharmacognosy. By T. E. Wallis. Published by S. K. Jain Publishers and distributors Pvt. Ltd., for CBS 5th, edition, London 2005.

❑ Web sites

Thankyoooo

For

your

listening